

Problem Set 2

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Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following tables was created using the data from a study run in a major Latin American city.¹ As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, “We can solve this the easy way” to draw a bribe). The table below shows the resulting data.

¹Fried, Lagunes, and Venkataramani (2010). “Corruption and Inequality at the Crossroad: A Multi-method Study of Bribery and Discrimination in Latin America. *Latin American Research Review*. 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

- (a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```

1 bribe_table <- as.table(matrix(c(14, 6, 7, 7, 7, 1),
2                               nrow = 2, byrow = TRUE,
3                               dimnames = list(
4                                 class = c("upper-class", "lower-class"),
5                                 party = c("not-stopped", "bribe-requested", "stopped-
6                                     warning")
7                               )))
8 row_totals <- rowSums(bribe_table)
9 col_totals <- colSums(bribe_table)
10 grand_total <- sum(bribe_table)
11 expected <- outer(row_totals, col_totals) / grand_total
12 dimnames(expected) <- dimnames(bribe_table)
13 chi_contrib <- (bribe_table - expected)^2 / expected
14 chi_square_stat <- sum(chi_contrib)

```

The X^2 value is 3.79. I did this by finding the row totals, col totals and grand totals. This was used to find the expected values. I then found the chi-square contribution for each cell and added up all the contributions to get the x^2 value.

- (b) Now calculate the p-value from the test statistic you just created (in R).² What do you conclude if $\alpha = 0.1$?

```

1 df <- (nrow(bribe_table) - 1) * (ncol(bribe_table) - 1)
2 p_value <- 1 - pchisq(chi_square_stat, df)

```

The p-value is 0.15 and since the p value is greater than 0.1 I fail to reject the null hypothesis. This means that there is no significant evidence that class and outcome are related.

²Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

```
1 std_residuals <- (bribe_table - expected) / sqrt(expected)
```

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.1360828	-0.8153742	0.818923
Lower class	-0.1825742	1.0939393	-1.098701

(d) How might the standardized residuals help you interpret the results?

The standardized residuals show that none of the differences between observed and expected counts are large, meaning class and outcome appear to be independent.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.³ Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: <https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv>

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

³Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

- (a) State a null and alternative (two-tailed) hypothesis.

H0= reservation policies do not change the number of new or repaired drinking water facilities in the villages. Ha= reservation policies do change the number of new or repaired drinking water facilities in the villages.

- (b) Run a bivariate regression to test this hypothesis in R (include your code!).

```
1 summary(lm(water ~ reserved, data = village_data))
2 #Coefficients:
3 # Estimate Std. Error t value Pr(>|t|)
4 #(Intercept) 14.738      2.286    6.446 4.22e-10 ***
5 # reserved    9.252      3.948    2.344 0.0197 *
6
7 #Residual standard error: 33.45 on 320 degrees of freedom
8 #Multiple R-squared: 0.01688, Adjusted R-squared: 0.0138
9 #F-statistic: 5.493 on 1 and 320 DF, p-value: 0.0197
```

- (c) Interpret the coefficient estimate for reservation policy.

The correlation coefficient is 9.255 for the reservation policy. For a one unit increase in reservation policy there is an increase in drinking water facilities of 9.25 units.